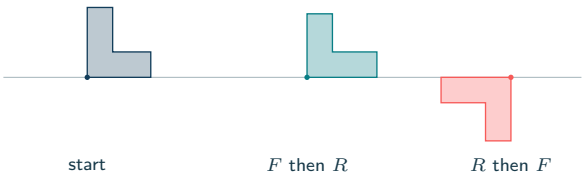
ALGEBRA LADDER 7 OF 9

Matrices: machines that do not commute

A matrix is a machine on the plane, the product of two matrices is one machine after another, and the order changes the machine.

1 / THE RULE AND THE PICTURE

Choose R for rotation, F for flip (reflection), and S for shear



Dots mark separate origins; the L-shapes are spaced for comparison.

Here R turns $90°$ anticlockwise about the origin; F reflects in the x-axis. Read products from right to left: RF means F first, then R.

$$\boxed{RF \neq FR}$$

RF reflects in $y = x$; FR reflects in $y = -x$. The two final shapes differ by a $180°$ turn.

2 / THREE MACHINES

The columns of a matrix say where the two unit steps go

name	machine	matrix	columns: where the unit steps go
R	turn $90°$ anticlockwise	$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$	$(1,0) \to (0,1)$, $(0,1) \to (-1,0)$
F	reflect in the x-axis	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	$(1,0) \to (1,0)$, $(0,1) \to (0,-1)$
S	shear to the right	$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$	$(1,0) \to (1,0)$, $(0,1) \to (1,1)$

To multiply, go along a row of the left matrix and down a column of the right one: RF has top row $(0 \cdot 1 + (-1) \cdot 0,\ 0 \cdot 0 + (-1)(-1)) = (0,1)$ and bottom row $(1 \cdot 1 + 0 \cdot 0,\ 1 \cdot 0 + 0 \cdot (-1)) = (1,0)$.

3 / WHICH SCHOOL LAWS STILL HOLD

Associativity and distributivity survive and four other habits fail

school law	holds for matrices?	example
associative	yes	$(RF)S = R(FS) = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$
distributive	yes	$R(F+S) = RF + RS = \begin{bmatrix} 0 & 0 \\ 2 & 1 \end{bmatrix}$
commutative	no	$RF = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ but $FR = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$
$(A+B)^2 = A^2 + 2AB + B^2$	no	$(R+F)^2 = 0$ but $R^2 + 2RF + F^2 = \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix}$
product zero forces a zero factor	no	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = 0$
cancel A from $AB = AC$	no	$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ gives $AB = AC$ for $B \neq C$

4 / WHAT THE FAILURES COST

Expanding, cancelling and solving take care

Because $AB \neq BA$ the expansion of $(A+B)^2$ has two middle terms $AB + BA$, and in the example they cancel exactly, leaving $(R+F)^2 = 0$ for two machines that are far from zero. Because a product can vanish with both factors non-zero, $AB = 0$ does not give $A = 0$ or $B = 0$, and an equation $AX = B$ can be solved by multiplying by the inverse only when one exists, which is when the determinant is not zero: $\det R = 1$, $\det F = -1$, $\det S = 1$. When an inverse does exist the undo rule of the composition sheet returns, $(AB)^{-1} = B^{-1}A^{-1}$.

5 / TRY THESE

Multiply in both orders and compare

(a) Find $FS = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$ and $SF = \begin{bmatrix} 1 & -1 \\ 0 & -1 \end{bmatrix}$, which differ. **(b)** Find two matrices that commute: S and $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ commute because both are shears. **(c)** Check that $R^4 = I$, since four quarter turns return every point.

ASIDE / FROM TRIANGLES TO MATRICES

The six moves of the triangle are six matrices, and the same table of products appears with r a turn by $120°$ and s a reflection.

LINKS TO OTHER SHEETS

The linear algebra ladder develops matrices as transformations and inverses; the next sheet asks how rotations in space behave.

Convention. R: $90°$ anticlockwise rotation about the origin; F: flip in the x-axis; S: shear. AB means B first; I is the identity.

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